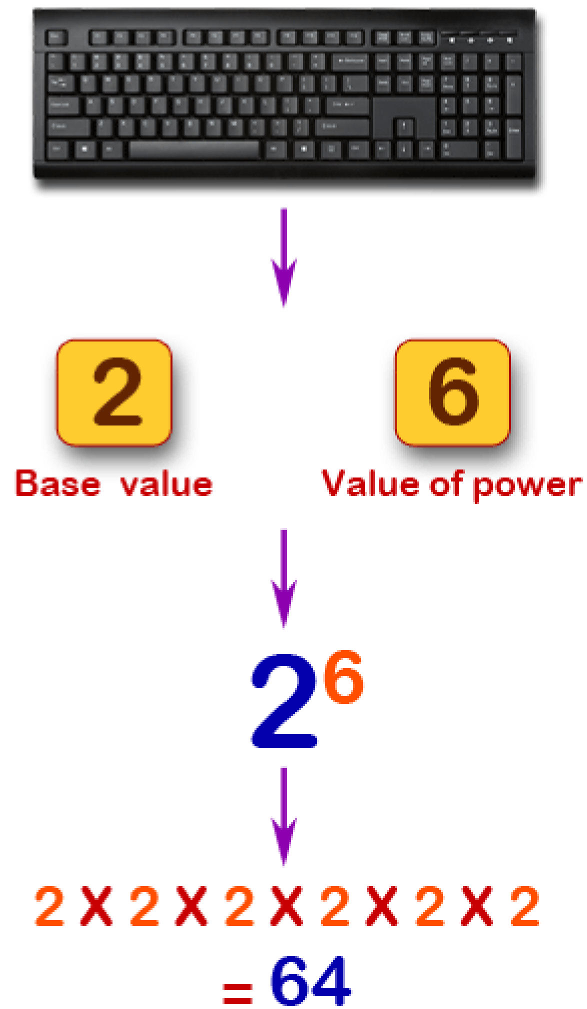
1. Power Calculation Recursion Variants

- Write a program in C to calculate the power of any number using recursion.
- Test Data :
- Input the base value : 2
  Input the value of power : 6
- *Expected Output* :
- The value of 2 to the power of 6 is : 64

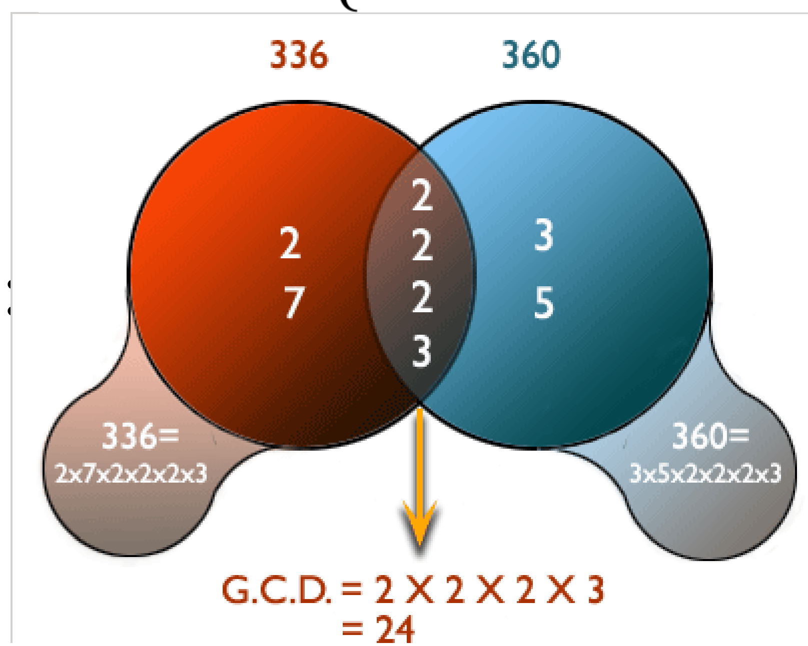


```c
#include <stdio.h>

long int CalcuOfPower(int x,int y)
{
    long int result=1;
    if(y == 0) return result;
    result=x*(CalcuOfPower(x,y-1));  //calling the  function CalcuOfPower itself recursively
}
int main()
{
    int bNum,pwr;
    long int result;
    printf("\n\n Recursion : Calculate the power of any number :\n");
    printf("----------------------------------------------------\n");

    printf(" Input the base  value : ");
    scanf("%d",&bNum);

    printf(" Input the value of power : ");
    scanf("%d",&pwr);

    result=CalcuOfPower(bNum,pwr);//called the function CalcuOfPower

    printf(" The value of %d to the power of %d is : %ld\n\n",bNum,pwr,result);

    return 0;
}
```

2. GCD Recursion Variants

- Write a program in C to find the GCD (Greatest Common Divisor) of two numbers using recursion.
- Test Data :
- Input 1st number: 10
  Input 2nd number: 50
- *Expected Output* :
- The GCD of 10 and 50 is:
- 10

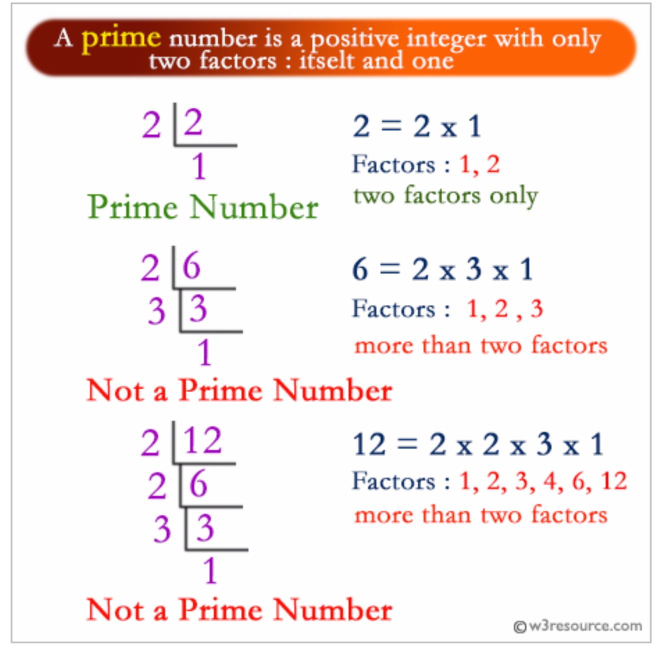
$$GCD(a,b) = \begin{cases} GCD(b, a-b) & \text{if } a > b \\ GCD(b, a) & \text{if } a < b \\ a & \text{if } a = b \end{cases}$$



```c
#include<stdio.h>

int findGCD(int num1,int num2);
int main()
{
  int num1,num2,gcd;
  printf("\n\n Recursion : Find GCD of two numbers :\n");
  printf("-------------------------------------------\n");
  printf(" Input 1st number: ");
  scanf("%d",&num1);
  printf(" Input 2nd number: ");
  scanf("%d",&num2);

  gcd = findGCD(num1,num2);
  printf("\n The GCD of %d and %d is: %d\n\n",num1,num2,gcd);
  return 0;
}

int findGCD(int a,int b)
{
     while(a!=b)
     {
          if(a>b)
              return findGCD(a-b,b);
          else
             return findGCD(a,b-a);
     }
     return a;
}
```

3 . Prime Check Recursion Variants

- Write a program in C to check if a number is a prime number or not using recursion.
- Test Data :
- Input any positive number : 7
- *Expected Output* :
- The number 7 is a prime number.



Prime number between 1 to 100

1	2	3	4	5	6	7	8	9	10
11	12	13	14	15	16	17	18	19	20
21	22	23	24	25	26	27	28	29	30
31	32	33	34	35	36	37	38	39	40
41	42	43	44	45	46	47	48	49	50
51	52	53	54	55	56	57	58	59	60
61	62	63	64	65	66	67	68	69	70
71	72	73	74	75	76	77	78	79	80
81	82	83	84	85	86	87	88	89	90
91	92	93	94	95	96	97	98	99	100

```c
#include<stdio.h>

int checkForPrime(int);
int i;

int main()
{

    int n1,primeNo;

    printf("\n\n Recursion : Check a number is prime number or not :\n");
    printf("--------------------------------------------------------\n");

    printf(" Input any positive number : ");
    scanf("%d",&n1);

    i = n1/2;

    primeNo = checkForPrime(n1);//call the function checkForPrime

   if(primeNo==1)
       printf(" The number %d is a prime number. \n\n",n1);
   else
      printf(" The number %d is not a prime number. \n\n",n1);
   return 0;
}

```

```
int checkForPrime(int n1)
{
    if(i==1)
    {
        return 1;
    }
    else if(n1 %i==0)
    {
         return 0;
    }
    else
       {
         i = i -1;
         checkForPrime(n1);//calling the function checkForPrime itself recursively
      }
}
```

4. Even/Odd Range Recursion Variants

- Write a program in C to print even or odd numbers in a given range using recursion.
- Test Data :
- Input the range to print starting from 1 : 10
- *Expected Output* :
- All even numbers from 1 to 10 are : 2 4 6 8 10
- All odd numbers from 1 to 10 are : 1 3 5 7 9

```c
#include <stdio.h>
void EvenAndOdd(int stVal, int n);

int main()
{
    int n;
    printf("\n\n Recursion : Print even or odd numbers in a given range :\n");
    printf("-------------------------------------------------------------\n");

    printf(" Input the range to print starting from 1 : ");
    scanf("%d", &n);

    printf("\n All even numbers from 1 to %d are : ", n);
    EvenAndOdd(2, n);//call the function EvenAndOdd for even numbers

    printf("\n\n All odd numbers from 1 to %d are : ", n);
    EvenAndOdd(1, n);// call the function EvenAndOdd for odd numbers
    printf("\n\n");

    return 0;
}
void EvenAndOdd(int stVal, int n)
{
    if(stVal > n)
        return;
    printf("%d  ", stVal);
    EvenAndOdd(stVal+2, n);//calling the function EvenAndOdd itself recursively
}
```